

- PDF Generation Module Design

The PDF generation module will transform processed web content into professionally formatted PDF documents with navigation, styling, and metadata.

1. PDF Document Structure

- Document Initialization: Create PDF document with configured page size/orientation
- Metadata Setup: Add title, author, creation date, keywords
- Page Configuration: Set margins, bleed, trim based on template
- Font Registration: Register and embed required fonts
- Style Definitions: Define global styles for headings, paragraphs, etc.

2. Cover Page Generation

- Template-Based Covers: Apply user-selected cover template
- Title Extraction: Use main page title or user-provided title
- Image Handling: Include site favicon or featured image if available
- Metadata Display: Show source URL, generation date, author if available
- Custom Branding: Add optional user branding or logo

3. Table of Contents Creation

- Hierarchy Detection: Identify document structure from headings
- TOC Generation: Create multi-level table of contents
- Page Numbering: Include accurate page numbers
- Linking: Make TOC entries clickable links to sections
- Styling: Apply template styling to TOC
- Dynamic Updates: Update TOC when content changes

4. Content Layout Engine

- Content Flow: Handle text flow across pages
- Page Breaks: Manage intelligent page breaks
- Column Layout: Support single or multi-column layouts
- Element Positioning: Position elements according to template
- Responsive Handling: Adapt content to page dimensions
- Orphan/Widow Control: Prevent single lines at top/bottom of pages

5. Element Rendering

- Text Rendering: Format paragraphs, headings, lists with proper styling
- Image Rendering: Insert images with proper scaling and resolution
- Table Rendering: Format tables with headers and proper page breaks
- Code Block Rendering: Maintain formatting and syntax highlighting
- Quote Formatting: Apply special styling to blockquotes
- List Formatting: Handle ordered and unordered lists

6. Navigation and References

- Bookmark Creation: Generate PDF bookmarks from headings
- Internal Linking: Convert internal links to PDF references
- External Links: Preserve external URLs as clickable links
- Cross-References: Handle figure/table references
- Page Numbers: Add configurable page numbering
- Headers/Footers: Add customizable headers and footers

7. Styling and Templating

- Template Application: Apply selected template to document
- Style Consistency: Ensure consistent styling throughout document

- Color Management: Handle color schemes with proper rendering
- Custom Styling: Apply user-defined styling preferences
- Dark Mode Support: Optional dark mode for PDF viewing
- Accessibility Features: Add accessibility tags and structure

8. Optimization and Finalization

- Image Optimization: Balance quality vs file size
- Font Subsetting: Embed only used characters from fonts
- Compression: Apply appropriate compression techniques
- File Size Management: Optimize for reasonable file sizes
- Compatibility Checks: Ensure PDF standards compliance
- Quality Assurance: Verify PDF structure and rendering

9. Preview Generation

- Lightweight Preview: Generate faster, lower-resolution preview
- Page Thumbnails: Create thumbnails for quick navigation
- Interactive Elements: Show interactive elements in preview
- Preview Caching: Cache preview for performance

10. Output Options

- Resolution Options: Offer different quality/size tradeoffs
- Format Variants: Support PDF/A for archiving if requested
- Security Features: Add optional password protection
- Watermarking: Apply optional watermarks
- Download Packaging: Prepare file for download with proper metadata

Technology Recommendations:

1. Core PDF Generation:

- React-PDF for React-based generation
- PDFKit for more complex document needs
- Puppeteer PDF generation for highest fidelity

2. Font Handling:

- Google Fonts integration
- Font fallback system
- Font subsetting for file size optimization

3. Image Processing:

- Sharp for server-side image optimization
- Canvas for client-side image handling

4. Templating System:

- Component-based templates
- CSS-in-JS for styling
- Theme variables for customization

5. Performance Considerations:

- Streaming generation for large documents
- Partial rendering for previews
- Background processing with update notifications